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Appendices

A Memory Ranking Mechanism of FINMEM

Upon receiving an investment inquiry, FINMEM retrieves the top- K critical memory events from each layer and channels them to the immediate reflection

component of the working memory. These events are selected based on their information retrieval score, γ_l^E , where l represents the layer (shallow, intermediate, or deep), as defined in Equation 2.

$$\gamma_l^E = S_{\text{Recency}_l}^E + S_{\text{Relevancy}_l}^E + S_{\text{Importance}_l}^E, \quad (2)$$

where each memory event is only associated with one score and can only belong to a single layer.

Let E denote a given memory event. The scoring mechanism for E , adapted from Park et al. (Park et al., 2023) but with modified recency and importance computations, is tailored to handle data with various timelines and to achieve layered processing that represents the diverse periodicities of the financial environment. This score encapsulates three metrics: recency (how recently the event occurred), relevancy (the event's pertinence to the current context), and importance (the event's significance). Individual metric scores exceeding 1.0 are scaled to the $[0,1]$ range before being summed, ensuring a balanced contribution from each component and preventing any single metric from dominating the overall score. The resulting composite score provides a comprehensive evaluation of the memory event's significance within the multi-layered, periodically varying financial landscape.

$$S_{\text{Recency}_l}^E = e^{-\frac{\delta^E}{Q_l}}, \quad \delta^E = t_P - t_E, \quad (3)$$

where δ^E represents the time elapsed between a memory event's occurrence and the trading inquiry's arrival. The model utilizes three processing layers, each corresponding to a specific timeframe: shallow ($Q_{\text{shallow}} = 14$ days), intermediate ($Q_{\text{intermediate}} = 90$ days), and deep ($Q_{\text{deep}} = 365$ days). These intervals represent two weeks, a quarter, and a year respectively.

When a trade inquiry P arrives in processing layer l via an LLM prompt, the agent calculates the recency score $S_{\text{Recency}_l}^E$ for a memory event E using Equation 3. This score inversely correlates with the time elapsed between the inquiry and the event's memory timestamp, mapping to Ebbinghaus's forgetting curve (Murre and Dros, 2015). The stability term Q_l in Equation 3 modulates memory decay rates across layers, with higher values in deeper layers indicating longer memory persistence. For instance, in the trading context, company annual re-

ports (e.g., Form 10-Ks) are assigned higher stability values and categorized within deeper processing layers compared to daily financial news, reflecting their extended timeliness, relevance, and impact on financial decision-making.

$$S_{\text{Relevancy}_l}^E = \frac{\mathbf{m}_E \cdot \mathbf{m}_P}{\|\mathbf{m}_E\|_2 \times \|\mathbf{m}_P\|_2} \quad (4)$$

The relevancy score $S_{\text{relevancy}_l}^E$ quantifies the semantic similarity between a memory event E and the current query P using cosine similarity of their respective embedding vectors, $\mathbf{m}_E$ and $\mathbf{m}_P$, as shown in Equation 4. These embeddings are generated from the event's textual content and the LLM prompt query (which includes trading inquiries and the agent's character setting) using OpenAI's "text-embedding-ada-003" model.

The importance score $S_{\text{Importance}_l}^E$ for a memory event E in layer l is calculated as the product of a value v_l^E (derived from a uniform piecewise scoring function, Equation 5) and a degrading ratio θ_l (Equation 6), as shown in Equation 7. This approach, adapted from (Park et al., 2023), is tailored to our stratified long-term memory structure. The likelihood of higher v_l^E values increases from shallow to deep layers, while θ_l measures the diminishing importance of an event over time using layer-specific exponential functions. The base α_l for each layer follows $\alpha_{\text{shallow}} < \alpha_{\text{intermediate}} < \alpha_{\text{deep}}$ (set to 0.9, 0.967, and 0.988 respectively), ensuring θ_l decreases to a threshold of 5 after 30, 90, and 365 days for shallow, intermediate, and deep layers. This layered approach, implemented through three-piece-wise functions for both $S_{\text{Importance}_l}^E$ and $S_{\text{Recency}_l}^E$, enables FinMem to process long-term memory in a stratified manner. Memory events are purged when $S_{\text{Recency}_l}^E$ falls below 0.05 or $S_{\text{Importance}_l}^E$ is under 5 (pre-scaling), maintaining the relevance and efficiency of the memory store.

$$v_l^E = \begin{cases} 40 & \text{with probability } p_1 \\ 60 & \text{with probability } p_2 \\ 80 & \text{with probability } p_3 \end{cases} \quad (5)$$

$$\theta_l = (\alpha_l)^{\delta^E}, \quad l = \text{shallow, intermediate, deep}, \quad (6)$$

where $p_1 + p_2 + p_3 = 1$, but their values vary by shallow, intermediate, and

deep processing. when shallow processing $p_1, p_2, p_3 = \{0.8, 0.15, 0.05\}$, intermediate processing, $p_1, p_2, p_3 = \{0.05, 0.8, 0.15\}$ and deep processing, $p_1, p_2, p_3 = \{0.05, 0.15, 0.8\}$.

$$S_{\text{Importance}_l}^E = v_l^E * \theta_l, \quad (7)$$

Furthermore, FINMEM employs an access counter function to dynamically manage memory events across layers, ensuring that crucial events influencing trading decisions are elevated to deeper layers for extended retention and recurring access. This process, monitored by the LLM validation tool Guardrails AI, tracks critical memory IDs across layers. Events deemed pivotal for investment success receive a 5-point boost to their importance score ($S_{\text{Importance}_l}^E$). Upon meeting upgrade criteria for a deeper layer, an event's recency score ($S_{\text{Recency}_l}^E$) is reset to 1.0, underscoring its significance and preventing rapid decay. Conversely, less relevant events gradually fade. This mechanism allows FINMEM to efficiently identify, prioritize, and retain key events based on their nature and retrieval frequency, while gradually phasing out less impactful information, thereby maintaining a dynamic and relevant memory structure for financial decision-making.

B Details on Evaluation Metrics

Below is a brief overview of these metrics:

Cumulative Return (CR) % measures the total value change of an investment over time by summing daily logarithmic returns, shown in Equation 8. Higher values indicate better strategy effectiveness.

$$\text{CR} = \sum_{t=1}^n r_i = \sum_{t=1}^n \left[\ln \left(\frac{p_{t+1}}{p_t} \right) \cdot \text{action}_t \right] \quad (8)$$

, where r_i is the logarithmic return from day t to $t + 1$, p_t and p_{t+1} are the closing prices on days t and $t + 1$, respectively, and action_t is the model's trading decision for day t .

Sharpe Ratio (SR) assesses risk-adjusted returns by dividing the average excess return (R_p) over the risk-free rate (R_f) by its volatility (σ_p), detailed in Equation 9. Higher ratios signify better performance.

$$\text{SR} = \frac{R_p - R_f}{\sigma_p} \quad (9)$$

Annualized Volatility (AV) % and Daily Volatility (DV) % quantify return fluctuations; AV is

derived by scaling DV (*standard deviation of daily logarithmic returns*) by the square root of the annual trading days (252), as in Equation 10. This metric highlights potential return deviations across the year.

$$\text{AV} = \text{DV} \times \sqrt{252} \quad (10)$$

Max Drawdown (MDD) % calculates the largest portfolio value drop from peak to trough, as given in Equation 11. Lower values indicate lesser risk and higher strategy robustness.

$$\text{MDD} = \max \left(\frac{P_{\text{peak}} - P_{\text{trough}}}{P_{\text{peak}}} \right) \quad (11)$$

Note that CR and the SR are often considered more essential than AV and MDD in evaluating asset trading performance due to their focus on long-term gains and risk-adjusted returns by their definition. Here, we regard these two metrics as primary metrics when evaluating the experiment outcomes.

C An example of mixed and lagged market signals: Partial investment insights of TSLA on 2021-03-05

Here are some insights from the agent's memory module for TSLA as of 2021-03-05. A few memory records are omitted; these are either neutral or positive. Despite this, the stock price trend for TSLA is sharply downward, conflicting with the overall positive financial sentiments and market signals. Utilizing proprietary models such as GPT4 and GPT-o1 as backbones, the financial decision-making agent can leverage other investment insights like historical momentum and self-reflection to consistently support a 'Sell' decision. In contrast, the large-sized open-source models like Qwen2.5-72B and DeepSeek-67B-Chat exhibit instability in producing consistent actions across repeated experimental trials.

Short-term Memory

- 1 **Sentiment: Negative:** The key insights from the news regarding Tesla Inc (NASDAQ: TSLA) losing market share to Ford Motor Company's (NYSE: F) Mustang Mach-E in the United States are as follows:1. ****Market Competition**:** Tesla is facing increased competition in the electric vehicle (EV) market, particularly from established automakers like Ford. The Mustang Mach-E's success indicates that other companies are effectively